

Refining Functional Vision Assessment: Challenges in Adapting Orientation and Mobility Tests to Virtual Reality

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ABSTRACT

The evaluation of functional vision in ophthalmology is challenging, often halting therapy development despite patient-reported Quality of Life (QoL) improvements. Orientation and Mobility (O&M) tests, which are difficult to be reproduced and costly due to their physical setup, can be enhanced using virtual reality (VR). This paper details a VR-based O&M test protocol that facilitates flexible design and access to extensive behavioral data. We addressed the challenge of creating an appropriate scoring system, identifying distinct error types and analyzing the contributing environmental and human factors. A refined scoring method was proposed, capturing the subtleties of functional vision beyond mere error counting. Our findings highlight the differences in behavior between VR scenarios and real-world settings, underscoring the importance of careful test design. This study offers new insights for future VR-based O&M test development and underscores the potential of VR in bridging the gap between virtual assessments and real-world functionality.

Index Terms: Functional Vision, Virtual Environment, Human Behavior, Navigation, Ophthalmology.

1 INTRODUCTION

Visual impairments are a growing issue in public health, significantly affecting individuals' Quality of Life (QoL) [12]. They can occur across all age groups [6], and pose important challenges in medical and social care. In order to address this problem, significant investments have been made, both in public research and in industry in order to develop new forms of therapies [11], prosthetics and remediation devices.

Despite the existence of functional vision assessment methods, enhancing the precision of these methods remains a challenge. The lack of objective assessment method prevents the continuation of several therapy development programs. There is a clear distinction between visual functions, which are features of the organ systems [8], such as visual acuity and contrast sensitivity, and functional vision, which refers to the ability to perform visual tasks in daily life. Thus, an evaluation of functional vision should be performed in real-world or close to real-world scenarios [3].

The Orientation and Mobility (O&M) test is one of the most objective test format available for assessing functional vision, as it evaluates participants' ability to navigate obstacles using visual cues [13]. However, these tests are limited by their physical nature, often being difficult to reproduce, costly to implement, and time-consuming for both patients and researchers. To overcome these

limitations, Virtual Reality (VR) has emerged as a promising alternative for O&M testing. VR enables the creation of reproducible, controlled environments with adjustable variables, while also providing access to richer behavioral data that is difficult to capture in physical tests. Previous studies have highlighted the potential of VR to replicate real-world navigation scenarios [1, 4]. Nonetheless, scoring metrics such as completion time and error counts may not fully adapt to the VR environment. For instance, some accidental collisions were reported in the first VR-based O&M test [4], yet they were still included in the scoring calculations. Also, the completion time is easily influenced by personal factors. Some visually impaired patients may complete the test quickly, while some normally sighted participants took longer to finish.

In this paper, we present our efforts to refine a specific VR-based O&M tests [9] by analyzing participants' behaviors during the test. A key focus of this work is on the scoring method, which has traditionally relied on the number of errors participants make during tasks. This approach, however, requires reconsideration in the context of VR. Through our analysis, we categorized these errors into distinct types and identified contributing factors, including environmental conditions (e.g., object contrast) and human factors (e.g., participants' familiarity with VR). These insights enabled us to refine the evaluation metrics, enhancing the accuracy and applicability of the VR-based functional vision assessment system.

Our findings suggest that while VR-based O&M systems address many limitations of physical tests, it is crucial to carefully consider behavioral differences between virtual and real-world scenarios. This study highlights the potential of VR in functional vision assessment and contributes to bridging the gap between virtual and real-world evaluations.

2 RELATED WORK

2.1 Functional vision evaluation

The most widely accepted objective assessment of functional vision is the Orientation and Mobility (O&M) test, as orientation and mobility are crucial skills for compensating for reduced visual information [13]. The state-of-the-art physical O&M test is the multi-luminance mobility test (MLMT) [6], which uses an obstacle course with varying light levels to track functional vision changes over time in subjects with inherited retinal dystrophies (IRDs). Although physical O&M tests can assess functional vision, they have several limitations. Reproducing physical environments is challenging, as some tests require large, dedicated spaces or controlled lighting conditions, making them difficult to implement in clinical settings. Additionally, physical tests can be time-consuming for both experimenters and participants, with setup changes and performance evaluation often relying on subjective scoring based on recorded videos or on-site observations. Furthermore, physical tests provide limited access to detailed behavioral data.

The emergence of VR technology offers new possibilities for O&M testing, as VR allows for greater flexibility in creating virtual

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environments and overcomes many of the limitations of physical tests. Virtual environments can be easily reproduced, and VR systems enable the collection of richer behavioral data through head-mounted display (HMD). Cogné et al. [7] reviewed the use of VR in diagnosing spatial navigation tasks, highlighting the potential for integrating VR technology into functional vision assessments. Some studies have already explored VR-based O&M tests. Aleman et al. [1] developed a VR O&M protocol based on the MLMT to evaluate functional vision in patients with IRDs, and Bennett et al. [4] later optimized this protocol by adding interactions to avoid accidental collisions. However, these previous VR O&M tests largely relied on the same evaluation metrics used in physical tests, without carefully considering the adaptability of these metrics to VR scenarios.

2.2 Evaluation scores used in O&M tests

Although O&M tests effectively assess quality of life, there is no gold standard for measuring O&M performance, nor a consensus on evaluation metrics [5]. In the MLMT [6], time and accuracy scores were based on course completion time and the number of collisions, a method that was similarly applied in its VR version [1], where Aleman et al. also measured the time before the participant took their first step. In the optimized version of the VR O&M test, as the task shifted from avoiding collisions to tagging objects, Bennett et al. replaced the number of collisions with the number of untagged objects as the primary metric. However, these evaluation metrics rely on statistical measures of performance, such as time and error counts, which may not fully adapted to virtual environments with different task settings. Therefore, in this study, we aimed to refine the evaluation metrics used in one VR-based O&M tests by refining the scoring method, and identifying the factors that influence performance in this VR scenarios.

3 METHODS

3.1 Virtual Reality-based O&M test system

To explore the usability and efficiency of integrating VR technologies into an O&M test system, we based our study on a protocol developed by Crozet et al. [9]. This protocol adapted the MLMT test [6] into a VR environment, and interactions with virtual environment were implemented. One major limitation of physical O&M tests is the potential danger posed by the required physical movement. This issue persisted in earlier VR-based O&M tests. This protocol adapted the real-walking displacement method used in physical tests into a torso-steering metaphor. It allows participants to remain seated throughout the test while controlling their avatar's direction by rotating their chair, with the avatar's speed fixed. This ensures the safety of participants and offers the option to stop the avatar at any time, providing greater control and comfort during the test. In this VR-based O&M test, the task was defined as "follow the path while destroying the obstacles you observe." To destroy an obstacle, participants used a virtual hand to interact with the object for 2 seconds, causing it to disappear, as illustrated in Figure 1. This task serves the same purpose as collision avoidance but is better suited to the VR environment, as it evaluates participants' ability to observe obstacles while avoiding accidental collisions through an active obstacle destruction mechanism. With this interactive design, the primary "errors" used for scoring were no longer based on physical collisions. Instead, errors were defined as "objects not destroyed by the end of the test," as missing an object indicates a failure to observe it.

The VR test system was designed with multiple adjustable environmental variables. Inspired by the MLMT test, six different luminosity levels can be selected for each test. Since visually impaired individuals retain spatial memory [2], eight distinct virtual labyrinths to counter potential biases introduced by memory effects (Two only for training, six only for tests) were designed. All

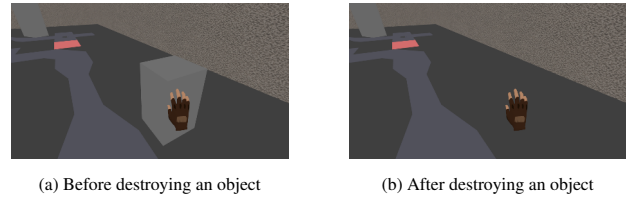


Figure 1: Object destruction mechanism.

labyrinths were of equal length and contained an identical number of objects placed at both high and low positions.

The VR-based O&M test system leverages a head-mounted display (HMD), enabling the capture of subtle participant movements throughout the test. This integration of VR technology provides us with the opportunity to gather rich behavioral data from various experiments, facilitating an in-depth analysis of the functional vision evaluation system.

3.2 Data collection

Using the VR-based O&M system, we conducted experiments to collect preliminary data from both healthy and visually impaired participants. Since the test protocol that we used is still in the development stage and patients have limited tolerance for extended testing, we conducted a pilot study with 41 normally sighted participants and 9 visually impaired patients with varying pathologies as shown in Table 1. The study aimed to establish the performance of healthy participants as a baseline for the VR-based O&M test and to evaluate the performance of visually impaired patients.

Patient	Gender	Age	Diagnostic
1	F	91	The patient has age-related macular degeneration, causing central vision loss in both eyes.
2	F	71	The patient has glaucoma and high myopia, resulting in both central and peripheral vision impairment in both eyes.
5	F	20	The patient has inflammatory optic neuropathy, leading to peripheral vision impairment in both eyes.
6	M	34	The patient is expected to behave like a healthy subject, with only one functional eye (the left eye, which functions normally).
7	M	30	The patient had a tumor on the visual pathways, resulting in impairment of only the temporal (outer half) visual field in both eyes.
8	M	42	The patient has hereditary optic neuropathy, causing both central and peripheral visual field impairment in both eyes.
9	M	61	The patient has orbital tumors on both sides, resulting in the loss of the lower visual field in both eyes.

Table 1: Diagnostic of patients

Each healthy participant completed twelve tests, while the patients conducted one or two tests. Throughout the experiments, data on behaviors such as head and hand movements, as well as avatar position, were recorded. The study was approved by the local ethics committee, with all participants providing written consent. This

preliminary data serves as a solid foundation for analyzing and verifying the ecological validity of the adapted system.

3.3 Data visualization

In this study, we focus on spatio-temporal data that captures real-time information, such as the avatar's position, body rotation, head, and hand movements. Using Unity, we reproduced the viewport seen by participants through the HMD during the test, primarily utilizing data on head position, rotation, and hand position. This visualization tool allowed us to recreate each test scenario accurately and view exactly what participants observed in their HMD, providing valuable insights for exploratory analysis of their interactions and behaviors.

4 ANALYSES RESULTS

4.1 Identification of error types

Since previous O&M tests primarily rely on error counts as a key metric, we first focused on examining the "errors" that occurred in our VR-based O&M test. Specifically, we reviewed participants' behaviors surrounding the "errors," defined as missed objects during the test.

In tests conducted by normally sighted participants, we identified various causes for these errors. Some objects were missed because they never appeared in the participant's Field of View (FOV); in other words, the participant did not notice them due to their focus being elsewhere. Some objects remained in view for a significant duration but were still overlooked; meaning that these objects are difficult to be noticed to some participants. Additionally, certain objects were missed despite being clearly observed and seemingly intended to be interacted with. Interestingly, similar patterns were observed in tests conducted by visually impaired participants. This finding is novel and has not been previously reported in physical or VR-based O&M tests. While VR technology enables us to track participants' real-time movements and FOV—allowing us to replay their test experiences—it also reveals discrepancies between behaviors in virtual and real-world environments.

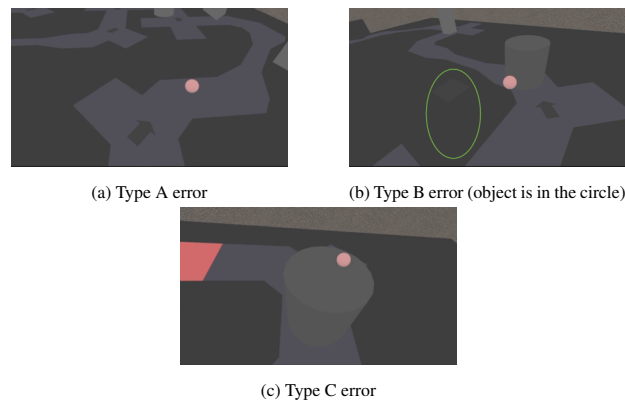


Figure 2: Different types of error from the viewport. We represented the participant's virtual hand with a small pink ball in the test environment.

To further investigate the characteristics of these errors, we analyzed data from moments surrounding missed objects. Specifically, we identified the key frame where the body position was closest to the missed object, then analyzed 500 frames before and 100 frames after this key moment, covering a total duration of 6.7 seconds at a 90 Hz sampling rate. Each observation was replayed in Unity and annotated by two experts who categorized the type of error. Three labels were assigned: Type A refers to "missed object is not

in FOV" as shown in Figure 2a, type B refers to "missed object is in FOV but not seen" as shown in Figure 2b, and type C refers to "missed object is seen" as shown in Figure 2c.

The statistics on the distribution of different error types are presented in Figure 3. Among the tests conducted by normally sighted participants (N=479), 147 missed objects were identified and annotated. Among them, 41 errors of type A, 97 errors of type B, and 8 errors of type C were reported, respectively. For visually impaired participants (N=12), 35 missed objects were reviewed and annotated. Among them, 15 errors of type A, 16 errors of type B, and 4 errors of type C were reported, respectively.

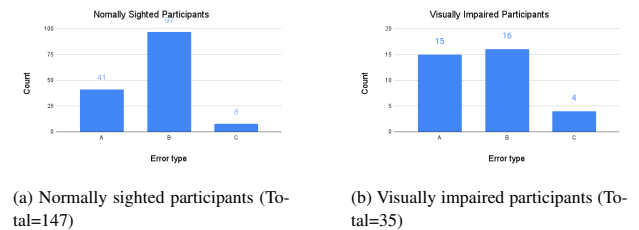


Figure 3: Frequency distribution of types.

4.2 Characteristics of refined errors

After annotating different error types, we sought to understand the underlying causes of these errors. We hypothesize that distinct factors contribute to each type of error. Leveraging the capabilities of VR, we extracted various configuration characteristics of missed objects within the virtual environment, including gray levels (represented by three identical RGB values in Unity), mean contrast across all luminosity levels, height, and volume. Since the errors made by normally sighted participants were not attributable to declines in functional vision, our initial analysis focused on the characteristics of missed objects in this group.

As illustrated in Figure 4, type A and type B errors exhibit distinct contrast values and heights: Errors annotated as type A are almost always in high positions, while errors annotated as type B predominantly have a low mean contrast value. Significance tests confirmed these differences: gray levels, treated as categorical variables, were analyzed using the Chi-Squared Test, while other characteristics were assessed with the Kruskal-Wallis test. Significant differences were identified in mean contrast values and heights (both with $P \ll 0.0001$). Post-hoc Dunn's tests further revealed that these differences primarily occurred between type A and type B errors (both with $P \ll 0.0001$).

Our findings suggest that type A errors are associated with objects positioned higher in the virtual environment, while type B errors are linked to low-contrast objects. In contrast, type C errors do not show specific patterns related to object characteristics. By combining these results with visualizations of behavioral data, we infer that type A errors likely result from insufficient exploration of upper areas, type B errors stem from difficulties in recognizing low-contrast objects, and type C errors may arise due to participants' unfamiliarity with the VR setup or unexpected technical issues.

Type A errors are interesting as they highlight the role of using peripheral vision in high position object searching tasks. Peripheral vision helps in identifying potential targets for saccades [10]. Type B errors are also noteworthy, as they involve low-contrast objects that were in the Field of View (FOV) but not seen. This type of error is significant for recognizing individuals who struggle with recognizing low-contrast objects. Type C errors, characterized by missed objects that were in the FOV and seen by participants, appear less discriminative for functional vision assessment. An analysis of normally sighted participants who made type C errors re-

Participant	Age	Experience of VR
2	62	Yes
6	47	No
11	22	No
12	22	No
16	31	Yes
22	20	Yes
24	55	Yes

Table 2: Demographic information of healthy participants who have made type C error.

vealed a pattern: these errors were more common among older participants (Mean age = 32) and younger participants with little to no VR experience, as summarized in Table 2. Additionally, technical issues may play a role, as evidenced by Participant 16, who reported problems with VR controller detection. This suggests that type C errors are primarily associated with media-related factors, such as technical limitations or user inexperience, rather than vision-related factors.

By analyzing the characteristics and underlying causes of each error type, these insights provide a basis for developing a more refined and robust scoring method tailored to the VR-based O&M test system.

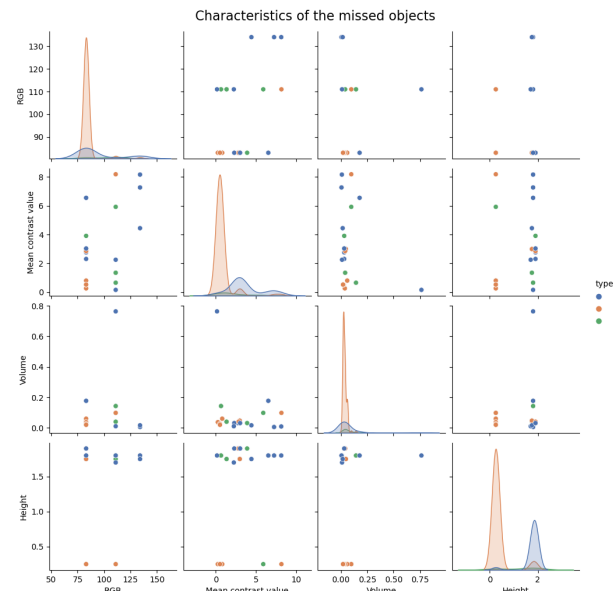
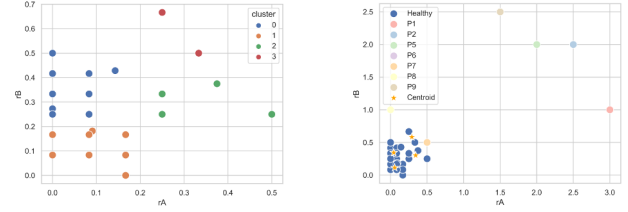


Figure 4: Characteristics of missed objects on healthy subjects' data.

4.3 Validation of refined scoring method

To leverage the exploratory results on the causes of errors during the test, we propose a refined scoring method based on the classification of three error types. As our analysis highlights, VR environments differ significantly from the real world, and including all error types—particularly those unrelated to functional vision—can introduce noise and bias into the scoring system. Our findings show that type C errors are often influenced by factors such as age, VR experience, or technical issues rather than functional vision, making them more valuable for improving the test system. Therefore, we remove type C errors to denoise the scores, while retaining type A and type B errors.



(a) Normally sighted participants and clusters

(b) Normally sighted and visually impaired participants

Figure 5: Refined scoring method.

We calculated the ratios r_A and r_B representing the occurrence of each error type relative to the number of tests conducted, to serve as refined scores. Using K-means clustering, four distinct clusters were identified within the group of healthy participants, as shown in Figure 5a. Each cluster corresponds to healthy but varying levels of vision capabilities:

- Cluster 1: This group (24/41 participants) exhibited the best O&M test performance, demonstrating strong abilities in recognizing low-contrast objects and exploring high peripheral environments.
- Cluster 0: This group (11/41 participants) showed some difficulty in recognizing low-contrast objects.
- Cluster 2: This group (4/41 participants) experienced similar difficulties with low-contrast object recognition as Cluster 0 but faced greater challenges in exploring high peripheral environments.
- Cluster 3: This group (2/41 participants) displayed the poorest performance among healthy participants, struggling with both object recognition and peripheral exploration.

Figure 5b also shows the r_A and r_B ratios for visually impaired participants. The centroids of the four clusters identified in the healthy participant group are marked with asterisks. While the separation between normally sighted and some visually impaired participants is evident, the r_A and r_B ratios keep healthy participants closely grouped, reflecting their relatively consistent performance.

This refined scoring method offers better explainability. The x-axis represents performance in high peripheral exploration, while the y-axis reflects the ability to recognize low-contrast objects. Verified by considering the diverse symptoms and severity levels of patients, our refined scores can not only appropriately locate different patients, but also explain the reasons for their distribution. For instance, Patient 6 and Patient 7 are near the boundary of healthy participants (Two points are overlapped in the Figure 5b). Patient 6, who has vision in only one normal eye, demonstrates behavior similar to a healthy participant. Patient 7, whose impairment affects only the outer half of the temporal visual field without impacting upper or contrast-related vision, also aligns closely with the healthy group. Their proximity to this cluster is consistent with their diagnostic profiles. Patient 8, although more distant from the healthy cluster, performs better than other patients. His visual field test results (Oculus test MD = 4.1 for the right eye) indicate minimal deficits, the best among all visually impaired participants. The ophthalmologist suggested that Patient 8 may compensate effectively with his healthier eye during visual tasks, explaining his relative closeness to the healthy group.

Assuming that the clusters found in healthy participants serve as a performance baseline, we calculated a general score for each

patient based on their distances to the centroids of these clusters, as defined in Equation 1.

$$s = \|dist_0 + dist_1 + dist_2 + dist_3\|_2 \quad (1)$$

Where $dist_0$, $dist_1$, $dist_2$ and $dist_3$ are the Euclidean distances between the patient's data point and each of the four cluster centroids. The general score s synthesizes each patient's overall functional vision evaluation, offering ophthalmologists a comprehensive understanding of the patient's performance.

All calculated general scores, along with the high peripheral and contrast-related vision evaluation rankings represented respectively by r_A , r_B for the patients, are presented in Figure 6. This visualization demonstrates that our scoring system effectively represents overall functional vision performance while also capturing subtle differences between patients in their contrast-related vision and high peripheral exploration abilities.

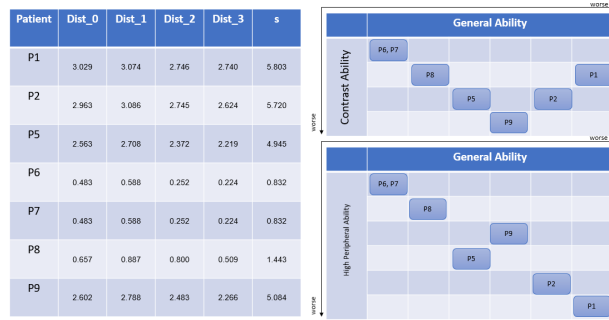


Figure 6: Scores and patients ranking.

5 CONCLUSION

Our study focuses on advancing the implementation of a VR-based O&M test system aimed at assessing functional vision. We analyzed data collected using a VR test protocol, which included data from both normally sighted and visually impaired participants. We identified three distinct types of errors within this VR-based O&M test system, which have not been previously discussed in previous tests. We then examined the characteristics of each error type and uncovered the contributing factors associated with them. These insights enabled us to refine the scoring method by incorporating the visual ability aspects reflected by each error type and using the performance of healthy participants as a baseline to calculate a general score for patients. Our refined scoring method provides a more reliable and interpretable measure of general functional vision as well as two specific aspects of functional vision, aligning closely with diagnostic results. Our findings highlight the significant potential of VR technology in ophthalmology, underscore the importance of addressing differences between virtual and real-world environments. This study establishes a robust foundation for both analyzing the existing VR-based O&M test system and developing future VR-based assessments. Future work will delve deeper into data analysis and explore the implementation of more realistic, dynamic scenarios for functional vision assessment.

REFERENCES

- [1] T. S. Aleman, A. J. Miller, K. H. Maguire, E. M. Aleman, L. W. Serano, K. B. O'Connor, E. C. Bedoukian, B. P. Leroy, A. M. Maguire, and J. Bennett. A virtual reality orientation and mobility test for inherited retinal degenerations: testing a proof-of-concept after gene therapy. *Clinical Ophthalmology*, pp. 939–952, 2021. 1, 2
- [2] E. M. Barhorst-Cates, K. M. Rand, and S. H. Creem-Regehr. The effects of restricted peripheral field-of-view on spatial learning while navigating. *PloS one*, 11(10):e0163785, 2016. 2

- [3] C. R. Bennett, P. J. Bex, C. M. Bauer, and L. B. Merabet. The assessment of visual function and functional vision. In *Seminars in pediatric neurology*, vol. 31, pp. 30–40. Elsevier, 2019. 1
- [4] J. Bennett, E. M. Aleman, K. H. Maguire, J. Nadelmann, M. L. Weber, W. M. Maguire, A. Maja, E. C. O'Neil, A. M. Maguire, A. J. Miller, et al. Optimization and validation of a virtual reality orientation and mobility test for inherited retinal degenerations. *Translational Vision Science & Technology*, 12(1):28–28, 2023. 1, 2
- [5] K.-y. J. Chang, L. L. Dillon, L. Deverell, M. Y. Boon, and L. Keay. Orientation and mobility outcome measures. *Clinical and Experimental Optometry*, 103(4):434–448, 2020. 2
- [6] D. C. Chung, S. McCague, Z.-F. Yu, S. Thill, J. DiStefano-Pappas, J. Bennett, D. Cross, K. Marshall, J. Wellman, and K. A. High. Novel mobility test to assess functional vision in patients with inherited retinal dystrophies. *Clinical & experimental ophthalmology*, 46(3):247–259, 2018. 1, 2
- [7] M. Cogné, M. Taillade, B. N'Kaoua, A. Tarruella, E. Klinger, F. Larue, H. Sauzéon, P.-A. Joseph, and E. Sorita. The contribution of virtual reality to the diagnosis of spatial navigation disorders and to the study of the role of navigational aids: A systematic literature review. *Annals of physical and rehabilitation medicine*, 60(3):164–176, 2017. 2
- [8] A. Colenbrander. Assessment of functional vision and its rehabilitation. *Acta ophthalmologica*, 88(2):163–173, 2010. 1
- [9] A. Crozet, L. Communier, T. Vigier, P. Lebranchu, and P. Le Callet. A virtual mobility test to evaluate functional vision of visual impaired patients. In *IMXw'23: ACM International Conference on Interactive Media Experiences Workshops*. ACM, 2023. 1, 2
- [10] C. J. Erkelens, I. T. C. Hooge, et al. The role of peripheral vision in visual search. *Journal of Videology*, 1:1–8, 1996. 3
- [11] A. M. Maguire, K. A. High, A. Auricchio, J. F. Wright, E. A. Pierce, F. Testa, F. Mingozzi, J. L. Bennicelli, G.-s. Ying, S. Rossi, et al. Age-dependent effects of rpe65 gene therapy for leber's congenital amaurosis: a phase 1 dose-escalation trial. *The Lancet*, 374(9701):1597–1605, 2009. 1
- [12] L. Schmetterer, H. Scholl, G. Garhöfer, L. Janeschitz-Kriegl, F. Corvi, S. R. Sadda, and F. A. Medeiros. Endpoints for clinical trials in ophthalmology. *Progress in Retinal and Eye Research*, 97:101160, 2023. 1
- [13] G. Virgili and G. Rubin. Orientation and mobility training for adults with low vision. *Cochrane Database of Systematic Reviews*, (5), 2010. 1